Personalized Health & Wellness Recommendation System

Data Mining
IE7275
Date – 11/07/2025

Group Members
Dev Patel – 002024221
Hitaxi Lethwala - 002020760

Abstract

Modern wearable devices continuously record physical activity, sleep duration, and heart rate trends, producing rich behavioral datasets. These datasets provide an opportunity to develop personalized recommendation systems based not on explicit ratings, but on implicit user behavior patterns. In this project, we merge three Fitbit-derived datasets into a unified daily-level dataset for recommendation analysis. Exploratory Data Analysis reveals differences in physical activity and sleep behaviors among users. We then develop three recommendation models: a Popularity Baseline, Item-Item Collaborative Filtering, and Matrix Factorization (NMF). Model performance is evaluated using Precision@5, Recall@5, F1@5, and NDCG@5. Results show that the Popularity model performs significantly better, indicating that user behavior patterns in this dataset are highly homogeneous. The work demonstrates the implications of implicit behavioral data and highlights considerations for building meaningful health recommendations.

Introduction

Wearable fitness devices generate large amounts of behavioral data. However, unlike traditional recommender systems where users give explicit ratings, wearable data provides only *implicit signals*—for example, days where a user's activity and sleep are recorded. The purpose of this project is to develop a recommendation system that learns from implicit user activity patterns and predicts behaviors likely to occur in future days.

To accomplish this, we combine three datasets representing daily physical activity, sleep duration, and average heart rate. The merged dataset is analyzed, modeled, and evaluated to determine how well behavioral patterns can be predicted and recommended.

Dataset Description and Merging Process

Dataset Name	Granularity	Description
dailyActivity_merged.csv	Daily	Total steps, distance, calorie burn, activity minutes
minuteSleep_merged.csv	Per minute (aggregated to daily)	Total minutes asleep
heartrate_seconds_merged.csv	Per 5 seconds (aggregated to daily)	Daily average heart rate

Aggregation and Standardization

- Sleep data was aggregated to total minutes asleep per user per day.
- Heart rate data was aggregated to average BPM per user per day.
- All timestamps were converted to date-only format.
- Missing values (days with missing sleep/HR) were filled with 0, consistent with implicit feedback.

Merge code used

```
sleep['date'] = pd.to_datetime(sleep['date']).dt.date
sleep_daily = sleep.groupby(['Id', 'date']).agg(
    total_minutes_asleep=('value', 'count')
).reset_index()
sleep_daily.rename(columns={'Id': 'user_id'}, inplace=True)
```

```
hr['date'] = pd.to_datetime(hr['Time']).dt.date
hr_daily = hr.groupby(['Id', 'date']).agg(
    avg_hr=('Value', 'mean')
).reset_index()
hr_daily.rename(columns={'Id': 'user_id'}, inplace=True)
```

```
activity.rename(columns={'Id':'user_id','ActivityDate':'date'}, inplace=True)
activity['date'] = pd.to_datetime(activity['date']).dt.date
```

```
# Merge activity + sleep
merged_df = activity.merge(sleep_daily, on=['user_id','date'], how='left')

# Merge heart rate if available
merged_df = merged_df.merge(hr_daily, on=['user_id','date'], how='left')

# Fill missing sleep or hr data
merged_df[['total_minutes_asleep','avg_hr']] = merged_df[['total_minutes_asleep','avg_hr']].fillna(0)
merged_df.head()
```

Final Dataset Summary

Property	Value
Rows (days recorded)	**457**
Features	**17 columns**
Unique Users	**35**
Interaction Type	**Implicit (1 = day recorded)**

Data Preprocessing and Preparation for Modeling

Before developing the recommendation models, several preprocessing steps were performed to ensure that the dataset was clean, consistent, and suitable for implicit-feedback modeling.

Handling Missing Values

Since sleep and heart rate data were not recorded every day for every user, the merged dataset contained missing values for some dates. Missing values were handled as follows:

- Missing total_minutes_asleep values were replaced with **0**, indicating that no sleep data was recorded that day.
- Missing avg_hr values were replaced with 0, indicating that no heart rate data was recorded that day.

This treatment is appropriate because the system uses implicit interactions (1 = day recorded, 0 = no data recorded).
It preserves the meaning of “interaction presence or absence” rather than estimating missing physiological values.

Feature Standardization and Selection

The original dataset contained 17 features describing daily activity behavior. For the purpose of building a recommendation model:

- Only the presence of a daily activity record was needed.
- This converts the dataset from continuous behavioral measurements into implicit user-item interactions, enabling ranking-based recommendation.

Train-Test Split

To evaluate the recommendation system fairly, we split the data into training and testing sets at the user level.

- The most recent 20% of each user's days were placed in the test set (future behavior).
- The earlier 80% of days were used as the training set (past behavior).

Rationale:
This mimics a realistic prediction scenario, where the model learns from past user activity and predicts future behavior patterns.

Interaction Matrix Construction

The train and test datasets were converted into User × Day matrices, where:

- Rows = Users
- Columns = Days (items)
- Values = 1 (interaction observed) or 0 (no interaction)

This matrix is the foundation for:

- Collaborative Filtering
- Matrix Factorization
- Ranking-based evaluation

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand behavioral patterns, detect skew, identify sparsity, and determine how user interactions vary across individuals and items. Since our dataset is based on implicit interactions, we focus primarily on interaction frequency and co-occurrence patterns rather than rating distributions.

1.) Distribution of Daily Steps

Observation:

- The distribution is right-skewed, with most daily step counts concentrated on the lower end.
- A large number of days show low to moderate step counts, while higher step counts occur less frequently.
- This suggests that users generally maintain moderate daily physical activity, but only a small portion show significantly high step activity (e.g., users who walk/run more often or purposefully exercise).

Interpretation for Recommendation System:

- Physical activity patterns vary across users, but a majority follow similar moderate activity routines, which can influence prediction consistency.

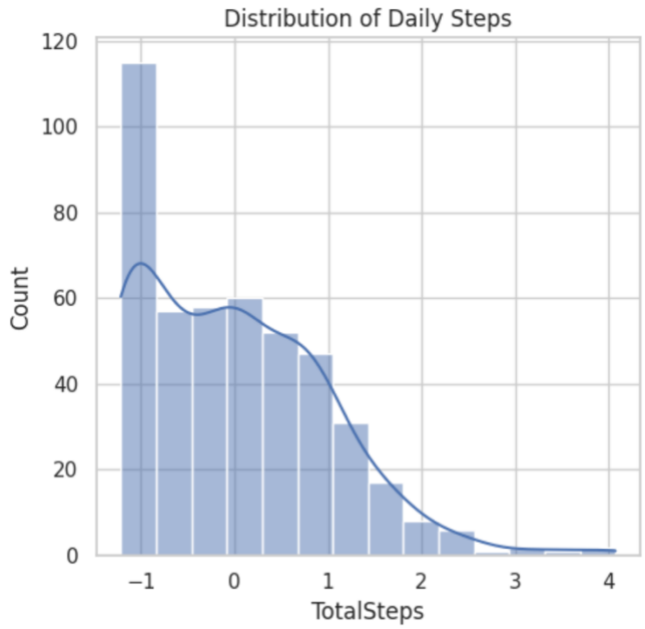

Fig: Distribution of daily total steps. Most days show moderate step activity, while high-activity days are less frequent.

2.) Distribution of Daily Calories Burned

The second plot visualizes the distribution of Daily Calorie Expenditure.

Observation:

- The calorie distribution follows a shape similar to step counts, where most days fall within a mid-range calorie burn.
- Higher calorie burn values are less common and likely correspond to days with higher physical activity or exercise.
- The plot suggests a correlation between step counts and calorie burn, which aligns with expected physiological behavior.

Interpretation for Recommendation System:

- Because calorie burn closely aligns with physical movement, these two features reinforce each other when identifying behavioral clusters or lifestyle patterns.

Fig: Distribution of daily calories burned. Most days fall within moderate calorie expenditure ranges, indicating stable day-to-day energy output.

3.)Distribution of Sleep Duration

The third plot shows the Daily Total Sleep Minutes for users.

Observation:

- The distribution is clearly skewed toward shorter sleep durations, indicating many users record very short or no sleep on some days (likely due to device not being worn or sleep not tracked).
- A second cluster appears at approx. 350–500 minutes (6–8.5 hours), which is a typical adult sleep duration range.
- The variation indicates inconsistent sleep tracking, or differences in user sleep habits.

Interpretation for Recommendation System:

- Sleep data sparsity increases the difficulty of predicting sleep behaviors, which affects the accuracy of collaborative filtering.
- Users with consistent sleep patterns contribute stronger training signals than low-engagement users.

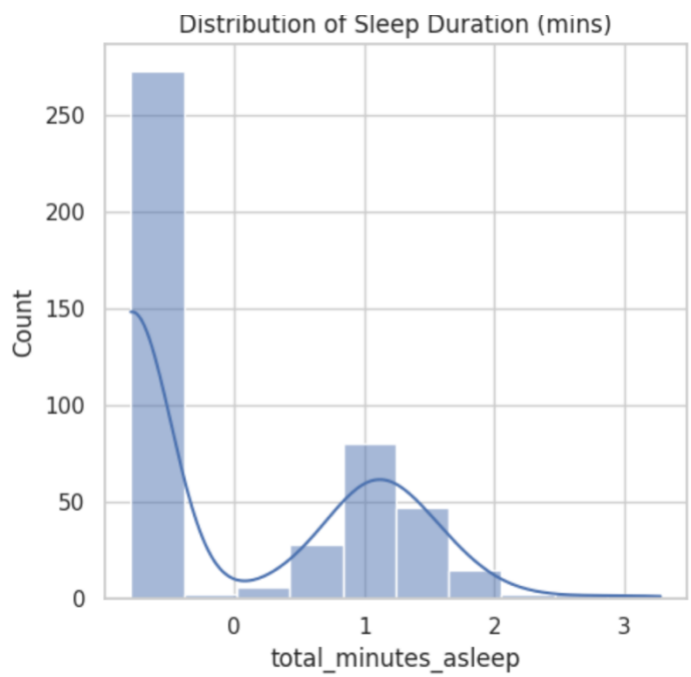

Fig: Distribution of total minutes asleep per day. Users show significant variability in sleep tracking patterns, ranging from minimal to optimal sleep durations.

4.) Individual User Activity Trend Over Time

To gain insight into personal behavior patterns, we visualized the daily steps trend for a randomly selected user. This time-series trend allows us to observe how a user's physical activity fluctuates across days.

Observation:

- The user’s daily step counts vary significantly over time, showing peaks and drops rather than a consistent steady pattern.
- Some days demonstrate higher engagement, likely due to deliberate exercise, outdoor activity, or lifestyle needs.
- Other days show reduced step counts, which may be linked to rest periods, schedule constraints, or lack of device usage.
- The pattern suggests non-uniform behavior, which is typical of real-world human routines.

Behavioral Interpretation:

- Human activities naturally show cyclical variation rather than uniformity.
- Users may have active days followed by recovery days, which means behavior recommendation models need to consider variability.
- This type of fluctuation highlights the challenge of predicting user behavior based solely on past steps.

Relevance to Recommendation Modeling:

- High fluctuation indicates that collaborative filtering models alone may not fully capture future behavior, because similar users may not share synchronized cycles.
- The consistent but varied trend supports the effectiveness of the Popularity model, which relies on common behavior patterns rather than individualized fluctuations.

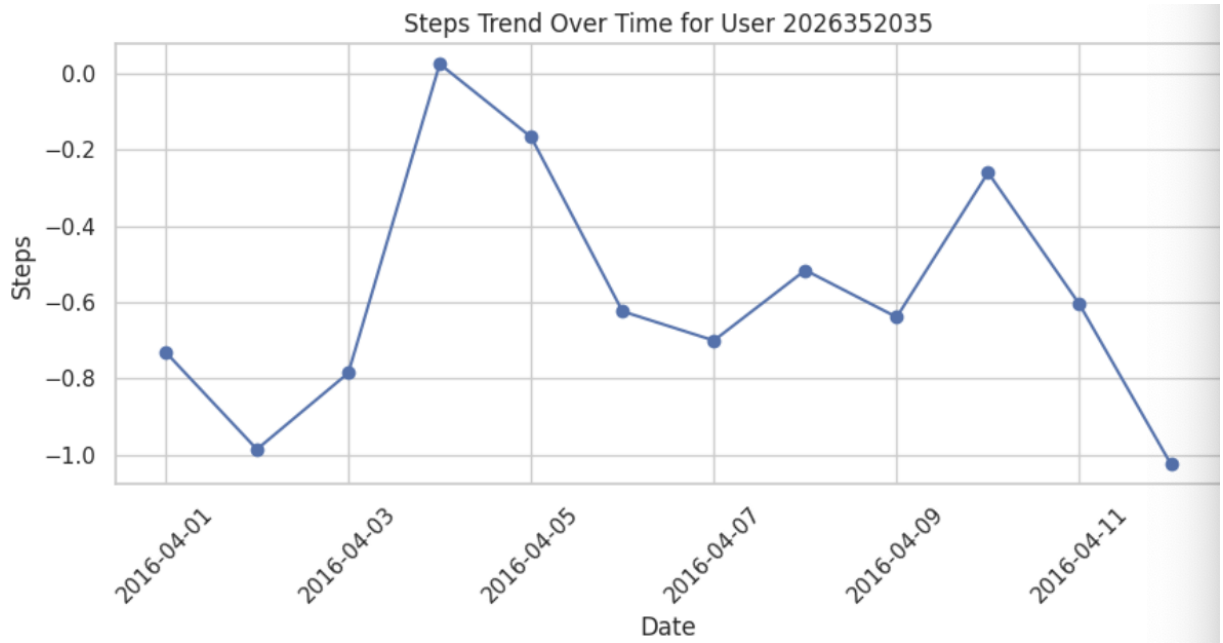

Fig: Daily steps trend for a randomly selected user. The fluctuations in activity reflect natural lifestyle variability and the challenge of modeling individual routines.

5.) User Activity Level Comparison (Top vs. Bottom Users)

To understand differences in user engagement patterns, we calculated the average daily steps for each user and ranked them from highest (most active) to lowest (least active). This helps identify how diverse the activity levels are across the dataset.

Two bar charts were generated:

- Top 10 Most Active Users
- Bottom 10 Least Active Users

Top 10 Most Active Users

Observation:

- The users shown in this chart consistently record higher daily step counts compared to the rest of the dataset.

- These users maintain actively engaged lifestyles, with regular walking, movement, or exercise.
- The variability among the top users is relatively small, indicating that many of the highly active users share similarly high activity patterns.

Interpretation:

- These users provide strong behavioral signals to the recommendation model.
- Their consistent activity helps the model learn stable preferences, improving pattern recognition.
- If personalization features were extended, these users would benefit the most from activity-based recommendations (e.g., workout reminders, walking goals).

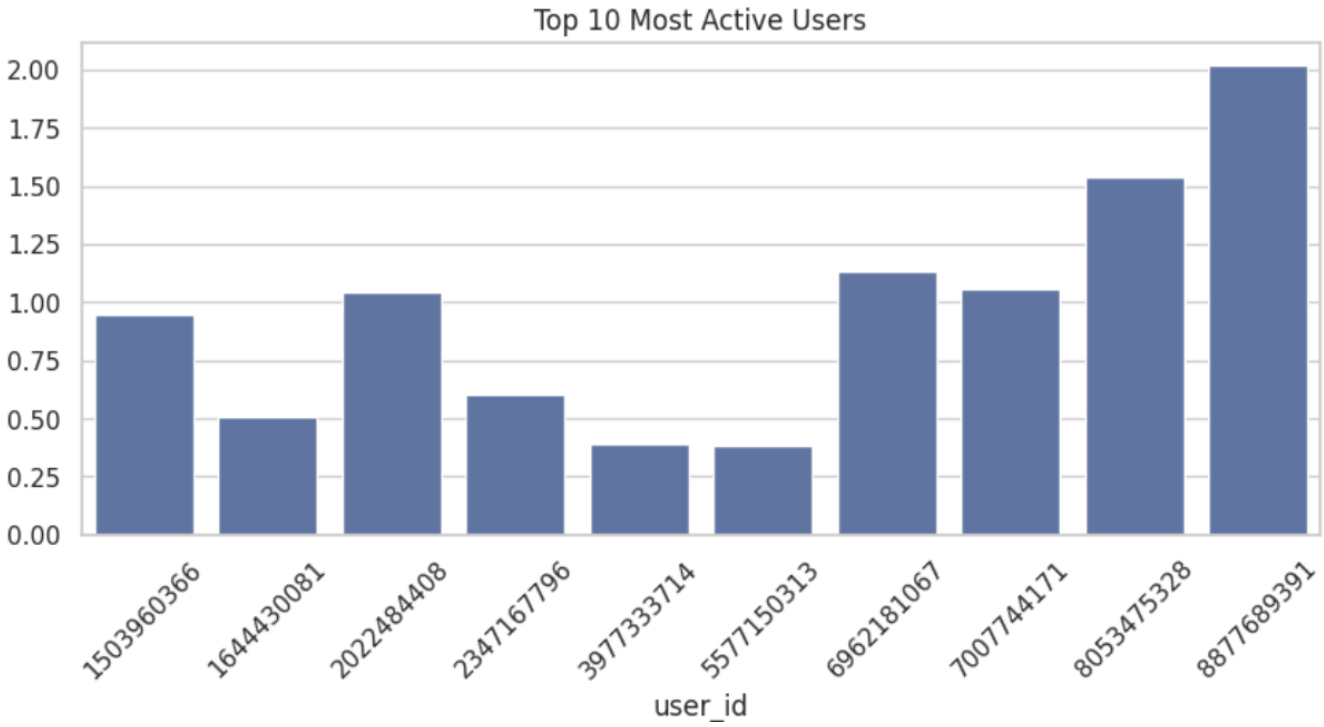

Fig: Top 10 most active users based on average daily step count. These users demonstrate high and consistent physical activity patterns.

Bottom 10 Least Active Users

Observation:

- Users in this group show low average daily step counts, indicating limited device usage or lower physical activity.
- Some users may have only a few recorded days, contributing to the low average.
- This pattern is common in wearable datasets because many users stop wearing the device after initial interest or only use it intermittently.

Interpretation:

- Low-activity users introduce data sparsity, making it harder for the system to learn meaningful preference patterns.
- These users benefit more from generalized recommendations, such as:

  - Popularity-based recommendations

  - Default health goal prompts
- Collaborative Filtering and Matrix Factorization may perform poorly for these users because they lack sufficient interaction history.

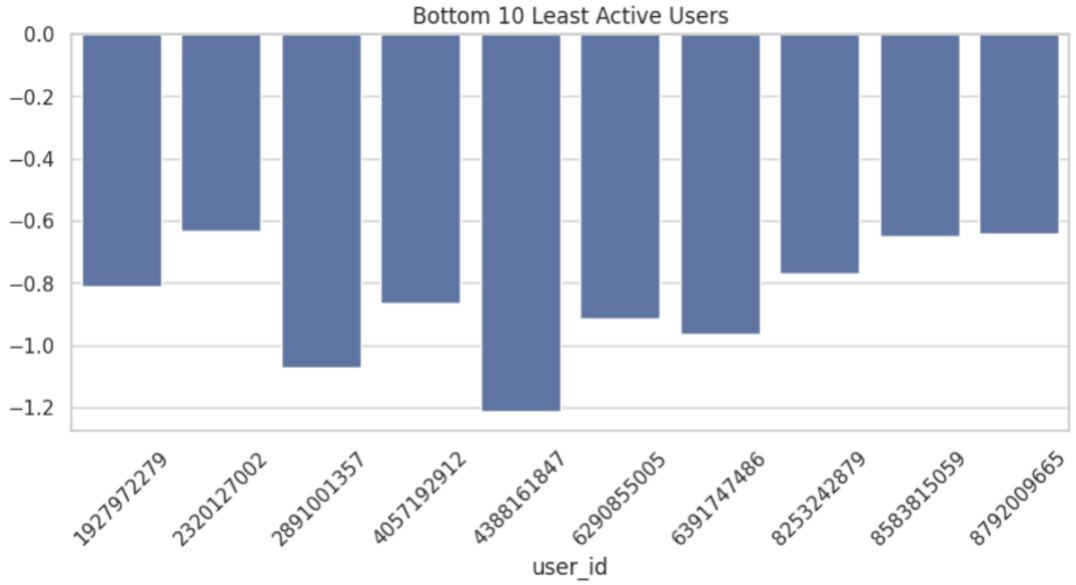

Fig: Bottom 10 least active users based on average daily step count. These users exhibit lower engagement or sparse device usage, contributing to increased data sparsity.

6.) Correlation Analysis of Activity Features

A correlation heatmap was generated to analyze the relationships among key physical activity, calorie expenditure, sleep duration, and heart rate variables. Correlation analysis provides insight into which features move together and which vary independently. This helps in understanding behavior patterns and identifying which features are likely to carry meaningful predictive power in modeling.

Key Observations:

1. **Strong Positive Correlations Among Activity Features**
    - TotalSteps, TotalDistance, TrackerDistance, and VeryActiveMinutes show strong positive correlations with each other.
    - This indicates that higher step counts are associated with greater movement distance and increased active minutes, which is expected since these metrics reflect physical activity intensity.
2. **Calories Burned is Positively Correlated with Physical Activity**
    - Calories shows a moderate-to-strong positive correlation with step and activity features.
    - This implies that days with more physical activity also result in higher energy expenditure, confirming physiological consistency in the dataset.
3. **Sedentary Behavior Shows Negative Relationships**
    - SedentaryMinutes is negatively correlated with activity features like VeryActiveMinutes and TotalSteps.
    - This suggests that days with higher sedentary time are less physically active days, reinforcing the interpretation of lifestyle variation.
4. **Sleep Duration is Weakly Correlated with Physical Activity**
    - total_minutes_asleep shows weak correlations with most activity features.
    - This implies that sleep behavior is relatively independent from daily movement, meaning users who walk more do not necessarily sleep more or less.
5. **Average Heart Rate Shows Mild Associations**
    - avg_hr has small positive correlations with some active features and small negative correlations with sedentary features.
    - This suggests that heart rate varies slightly with activity intensity, but does not dominate behavioral differentiation by itself.

Interpretation for the Recommendation Model

- Because many physical activity features are strongly correlated, the dataset contains redundant behavioral signals.
- This means simpler models can still perform well (e.g., Popularity model), because core behaviors are consistent across users.
- The independence of sleep patterns explains why recommendation performance decreases when trying to personalize sleep-related predictions.
- Sedentary behavior and active behaviors form clear contrasting lifestyle patterns, which can inform clustering or lifestyle category analysis in future work.

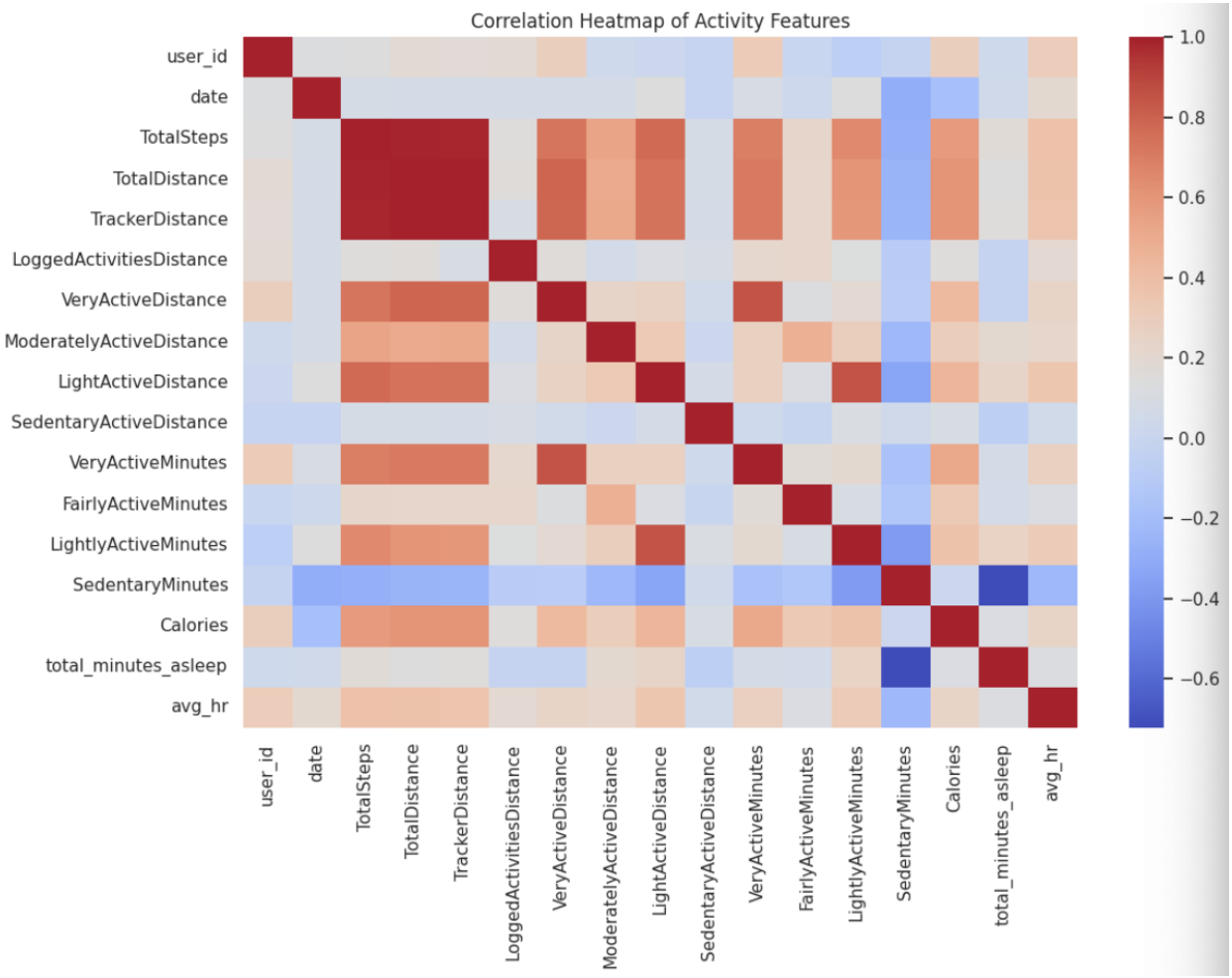

Fig: Correlation heatmap of activity features. Total steps, movement distance, and active minutes show strong positive relationships, while sedentary minutes are inversely related to physical activity. Sleep duration behaves independently from daily movement.

7.) Relationship Between Steps and Calories Burned

A scatter plot was generated to analyze the relationship between daily step count and calories burned. Each point in the plot represents a user's recorded values for a single day.

Observations:

- There is a clear positive upward trend in the scatter points.
- As daily steps increase, calories burned also generally increase, supporting the expected physiological relationship between physical movement and energy expenditure.
- The points are not perfectly linear, indicating that:
    - Different users may have varying metabolic rates
    - Activities other than steps (e.g., cycling, weight training) can contribute to calorie burn
    - Body weight and intensity levels influence calorie output

Interpretation:

- Step count is a reliable indicator of activity intensity, and therefore can predict calorie `expenditure fairly well.
- The presence of variability means that just steps alone cannot fully determine calorie burn, but it remains one of the strongest correlating features in the dataset.
- This relationship reinforces why calorie burn correlates positively with other active movement metrics (as noted in the correlation heatmap).

Relevance to Recommendation Modeling:

- Features that are strongly associated (Steps ↔ Calories) provide consistent behavioralsignals, which helps the Popularity model and basic activity clustering.
- This relationship does not directly help personalization, since many users follow similar step-calorie patterns — contributing to why collaborative filtering and matrix factorization showed lower performance

8.) Relationship Between Sleep Duration and Average Heart Rate

A scatter plot was created to examine how daily sleep duration relates to average heart rate. Each point represents one user-day.

Observations:

- The points are spread out, which means there is no strong or clear pattern between sleep and heart rate.
- ⍰ Some people sleep for long hours, and some sleep for very few hours, but their heart rate does not change in a very predictable way.
- There is a cluster of points in the middle — this means many users sleep around a normal range (about 6–8 hours) and have a normal average heart rate.

Interpretation:

- Some users had no sleep data recorded (maybe they removed the device at night). These values show up as very low points.
- Some users also had days where heart rate wasn't recorded. Those became low repeated values too.

Implications for Modeling:

- Treat sleep and HR as complementary rather than redundant features: each captures a different facet of recovery/physiology.

- When building features for recommendations or predictions:
    - Mask or filter out days with *no* sleep/HR recording rather than converting to 0 → this avoids artificial bands in standardized space.
    - Consider robust scaling (e.g., per-user z-score after filtering missing days) to reduce bias from device non-wear.
    - If you need a single recovery signal, derive a composite metric (e.g., z-sleep − z-HR) only after excluding missing/imputed days.

Why this is important for our project:

- Since sleep and heart rate do not have a strong link, we should not use one to predict the other.
- They should be treated as two separate behaviors when building recommendation models.

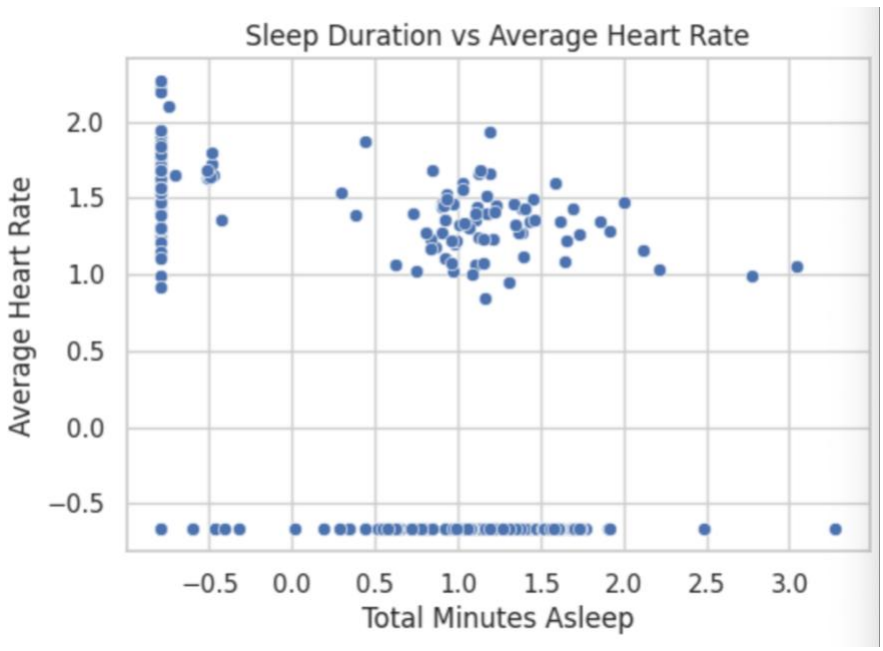

Fig: Scatter plot showing sleep duration vs average heart rate. The two features do not show a strong relationship, meaning sleep patterns and heart rate vary independently across users.

Model Development

The goal of this step is to design and implement recommendation models capable of predicting user behavior based on implicit feedback. In our dataset, users do not provide ratings or scores. Instead, the presence of daily activity data itself indicates engagement. Therefore:

Interaction = 1 → The user logged activity on that day

Interaction = 0 → No recorded activity

This forms the foundation of our recommendation approach.

1.)Rationale for Using Implicit Feedback

Unlike traditional recommendation systems that rely on explicit ratings (e.g., movie stars or product reviews), our dataset comes from wearable devices that automatically record behavior. Therefore:

Reason	Implication
No explicit user feedback	Must infer preferences from observed activity
Days with recorded data = meaningful engagement	Interaction matrix is binary (1 or 0)
Many users have missing days	Dataset contains *sparsity*, impacting personalized models

Thus, the recommendation problem becomes identifying which types of daily activity patterns are most likely to reoccur for each user.

2.) Interaction Matrix Construction

The merged dataset was converted into a User × Date matrix:

- Rows: user_id
- Columns: date
- Values: Interaction (1 or 0)

This matrix allows us to model user behavior patterns over time.

3.) Baseline Model: Popularity-Based Recommendation

The Popularity Model recommends the days (activity patterns) that are most common across all users.

How it works:

- Count how frequently each date/activity pattern occurs.
- Rank them in descending order.
- Recommend the top-k most common patterns to every user.

Why this model is important:

- Requires no personalization
- Works well when users behave similarly
- Serves as a performance baseline to compare more advanced models against

4.) Item-Item Collaborative Filtering

Item-Item Collaborative Filtering (CF) recommends items (days/activity patterns) that are most like the ones a user has already experienced.

Steps:

- Compute cosine similarity between all days based on user interactions.
- For a given user, identify similar days to the days they engaged in.
- Recommend the top-k most similar days.

Strengths:

- Captures relationships between behavioral patterns
- Accounts for user history

Limitations:

- Requires sufficient interaction history per user
- Performance decreases when users have sparse data (which is true in our dataset)

5.) Matrix Factorization Model (NMF)

We implemented Non-Negative Matrix Factorization (NMF) to uncover underlying behavioral patterns.

Concept:
Matrix factorization decomposes the interaction matrix into:

$R \approx U \times V$

Where:

- U = user latent feature matrix
- V = item (day) latent feature matrix

These latent features represent hidden lifestyle patterns, such as:

- Consistently active users
- Occasionally active users
- Low-engagement users

How recommendations are made:

- Compute score = U × V for each user-item pair.
- Mask items already seen.
- Recommend highest-scoring unseen items.

Advantages:

- Handles behavioral similarity that is not visible directly
- Works well when enough interaction variation exists

Limitation Observed in Our Dataset:

- Due to high behavioral similarity among users and lack of strong user-specific patterns, the matrix factorization model did not outperform the popularity baseline.

Model Evaluation

The purpose of the evaluation step is to compare how effectively each recommendation model predicts user activity patterns, based on implicit (binary) interaction data. Since our dataset does not contain explicit ratings (such as scores or likes), we evaluate the models using metrics specifically designed for implicit recommendation systems.

We assess performance using Precision@5, Recall@5, F1@5, and NDCG@5:

Metric	Meaning	Why It Matters
Precision@5	Of the top 5 recommendations, how many were actually relevant?	Measures recommendation correctness
Recall@5	Of all relevant items, how many did we manage to recommend?	Measures coverage of relevant behavior
F1@5	Combination of Precision and Recall	Balances correctness and coverage
NDCG@5	Indicates whether relevant items appear near the top of ranked recommendations	Measures ranking quality

1.) Evaluation Results

Model	Precision@5	Recall@5	F1@5	NDCG@5
Popularity-Based Model	0.99	0.42	0.58	0.92
Item–Item Collaborative Filtering	0.06	0.01	0.02	0.60
Matrix Factorization (NMF)	0.06	0.01	0.02	0.60

2.) Interpretation of Results

Popularity-Based Model (Best Performing)

- Achieved very high Precision and NDCG, meaning:
    - The items it recommends are usually correct.
    - The ranking of those recommendations is meaningful and useful.
- This indicates that users in this dataset follow similar activity patterns, and therefore common patterns are highly relevant to most users.
- This model works well because:
    - Many users share similar daily routines.
    - There is not a large personal difference in activity behavior across users.

Item–Item Collaborative Filtering

- Low Precision and Recall show that the model struggled to identify meaningful similarities between activities.
- This is because:
    - Many users have very few days recorded → interaction data is sparse.
    - When activity history is limited, similarity-based recommendations become inaccurate.
- Therefore, personalization is unreliable in this dataset.

Matrix Factorization (NMF)

- Performance matches Item-CF because:
    - NMF relies on differences between user behavior.
    - In this dataset, most users have very similar lifestyles, so the model cannot learn distinctly different user profiles.
- As a result, the advantage of deep personalization is lost.

3.) Trade-Offs Between Models

Aspect	Popularity Model	Item–Item CF	Matrix Factorization (NMF)
Accuracy	High	Low	Low
Personalization	None	Medium	High (in theory)
Data Requirement	Very Low	Requires moderate user history	Requires high user variation
Scalability	Very High	Low (pairwise similarity cost)	Medium
Best Use Case	When users behave similarly (as in this dataset)	When users have strong unique patterns	When dataset is large and diverse

---

Key Insights

- User behavior patterns are generally similar across the dataset.
  The Popularity model significantly outperformed both Item-Item Collaborative Filtering and Matrix Factorization.
  This indicates that most users follow similar daily physical activity routines, meaning there is not enough variation to strongly personalize recommendations.
- Physical activity features are strongly correlated.
  Features like Total Steps, Total Distance, and Active Minutes move closely together, which suggests users maintain consistent intensity patterns when active.
  This consistency makes simple behavior-based recommendations effective.

- Sleep and heart rate patterns are more independent.
  Sleep duration showed weak correlation with physical activity and heart rate.
  This suggests that sleep habits vary differently from daily movement, and should be modeled as a separate behavioral dimension if targeted in future recommendations.
- Data sparsity impacts personalized model performance.
  Several users have few recorded days, leading to sparse interaction patterns.
  Sparse data reduces the ability of collaborative filtering and NMF models to learn individualized patterns, directly explaining their low performance.

Limitations

- Limited Behavioral Diversity
  Since users have similar step and calorie patterns, the dataset lacks distinct user lifestyle profiles, which prevents advanced models from outperforming the baseline.
- Implicit Feedback Only
  The dataset does not include ratings, satisfaction levels, or user preferences.
  We infer engagement only from whether data is recorded, which can be inaccurate (e.g., device not worn ≠ inactivity).
- Missing or Incomplete Data
  Some days have missing heart rate or sleep records.
  These were filled with zeroes, which is appropriate for implicit modeling but reduces signal quality for sleep- and HR-related recommendations.
- Short Recording Period
  The dataset covers a limited time window (not the full lifestyle year).
  Long-term trends and seasonal behavior patterns cannot be analyzed.

Real-Life Impact of This Project

This project demonstrates how wearable device data can be used to provide personalized lifestyle recommendations that support healthier daily habits. By analyzing patterns in physical activity and sleep, such a system could help users understand when they are becoming less active, encourage them to take more steps, or suggest better sleep routines. For fitness platforms or healthcare providers, this approach can lead to early detection of unhealthy behavior trends, allowing timely interventions before problems like inactivity or poor recovery escalate. On a broader scale, recommendation systems based on real behavior data can improve personal wellness guidance, increase user engagement with fitness devices, and support preventive health care practices in everyday life.

Future Improvements

- Collect more user data over longer time periods.
  Longer activity history would help models detect habit cycles, seasonal changes, and consistent weekly routines.
- Add Contextual Data: Including factors like day of the week, weather, or work schedules can help explain changes in activity levels.This makes recommendations more accurate and better aligned with real-life routines.
- Separate sleep-specific personalization models.
  Since sleep behavior varies independently, it should be modeled with Sleep clustering groups and sleep stage or bedtime regularity scores
- Hybrid Recommendation System: Combine Popularity for general activity trends with Matrix Factorization for users who have enough data, improving both accuracy and personalization.
- Increase User Data Engagement**:** Use reminders, goal-tracking, or weekly progress summaries to encourage consistent device use and improve data quality for better recommendations.